

Demonstrating DoS Attacks: SYN Flood

Experiment: SYN Flood Attack Using Hping3

Aim: To provide an educational demonstration of a SYN flood DoS attack using Hping3 in a controlled environment, offering hands-on experience in identifying network vulnerabilities and understanding the impact of SYN flood attacks on system resources.

Objectives:

- To explain the mechanics of SYN flood attacks and their effects on network resources.
- To guide the use of Hping3 and nmap for port scanning and attack execution.
- To demonstrate traffic analysis with Wireshark to identify SYN flood patterns.
- To reinforce ethical hacking principles by conducting attacks in a secure, authorized setting.
- To increase awareness of DoS attack defenses and mitigation strategies.

Requirements:

- **Victim Machine:** A machine running Ubuntu Linux, configured as the target system for the attack.
- **Attacker Machine:** A machine running Kali Linux, equipped with tools necessary for launching the attack.
- **Tools:** Hping3 for attack deployment, nmap for port scanning, and Wireshark for traffic analysis.

Step 1: Identify the IP Addresses of the Victim and Attacker Machines

To successfully execute the attack, the IP addresses of both the attacker and victim machines must be noted.

On the Victim Machine (Ubuntu)

1. Open the terminal on the Ubuntu machine.
2. Enter the command:

Or ***Ifconfig***
ip a

3. Locate and write down the machine's IP address, which typically appears under inet in the output (e.g., 192.168.x.x).

On the Attacker Machine (Kali Linux)

1. Open the terminal on the Kali machine.
2. Enter the command:

ifconfig

3. Write down the IP address of the attacker machine for later use.

Step 2: Scan the Victim Machine for Open Ports

Verifying open ports on the victim machine is important to determine which ports are accessible. Often, certain ports (e.g., HTTP or HTTPS) are targeted in SYN flood attacks.

1. On the Attacker Machine in the terminal, type:

nmap [Victim's IP Address]

2. Review the output to confirm that necessary ports (such as 80 for HTTP) are open on the victim machine. IF NOT OPEN, OPEN IT FOR SIMULATION USING FOLLOWING SET OF CODE

```
To open ports 80:  
Sudo apt update  
Sudo apt install nginx  
Sudo ufw status  
Sudo ufw allow 80/tcp  
Sudo ufw reload  
Nmap <ip add>
```

Step 3: Configure the SYN Flood Attack Using Hping3

The Hping3 tool is a versatile command-line utility used for network testing and attack simulation. For this demonstration, it will be configured to flood the victim machine with TCP SYN packets.

1. In the terminal on the Attacker Machine, prepare the Hping3 command to simulate the flood.
2. The syntax for launching a SYN flood attack on a specific port (e.g., 80) with a

high packet size is as follows:

sudo hping3 -S -p 80 --flood [Victim's IP Address]

-S: Sets the SYN flag in each packet.

-p 80: Targets port 80 (HTTP port).

--flood: Sends packets at maximum speed.

3. Replace [Victim's IP Address] with the actual IP of the victim machine.

Step 4: Execute the SYN Flood Attack

Once configured, execute the attack by entering the command above. This will begin sending a high volume of TCP SYN packets to the victim's IP on the specified port.

Step 5: Monitor System and Network Impact

Observe the attack's effects by monitoring resource usage on the victim machine and analyzing network traffic on the attacker machine.

On the Victim Machine

1. Open the System Monitor tool in Ubuntu to observe resource utilization:

gnome-system-monitor

2. Look for significant increases in CPU and network usage as the machine processes large numbers of TCP SYN packets

On the Attacker Machine

1. Open Wireshark to capture and analyze the network traffic generated by the flood:

wireshark

2. Set filters to isolate TCP SYN packets or view traffic between the attacker and victim machines. Notice the pattern of repeated SYN requests, indicating the flood's effect on the target.

Conclusion and Clean-Up:

The DOS attack demonstration is complete. Ensure that you:

- Terminate the attack by stopping the Hping3 command by pressing CTRL + C
- Analyze and document any observed results.
- Restore normal operation on both machines by restarting network services if needed.

This exercise has shown how a SYN flood attack can disrupt system resources, providing valuable insights into network defense and traffic analysis for mitigation strategies.

Remember: Always respect ethical boundaries and use this knowledge to improve network security, not to harm or disrupt others.